

Boundary Motif Algebra (Structural Edition)

Adrian Fry

September 2026

Abstract

Boundary Motif Algebra formalises how structural motifs behave at boundaries. It defines boundary alignment, boundary interference, motif-boundary coupling, boundary collapse, boundary drift, and algebraic motif interactions across identity, horizon, collapse, and recurrence boundaries. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.

1 Introduction

Boundary Motif Algebra describes how motifs interact with structural boundaries. It governs:

- Boundary Alignment
- Boundary Interference
- Motif-Boundary Coupling
- Boundary Collapse
- Boundary Drift
- Algebraic Motif Interaction
- Boundary Regime Transitions

2 Boundary Alignment

Alignment condition:

$$B(M) > B_{\min}.$$

Alignment radius:

$$A = f(B(M)).$$

3 Boundary Interference

Interference strength:

$$I = h(M_1, M_2, d).$$

4 Motif-Boundary Coupling

Coupling strength:

$$C = g(M, B).$$

5 Boundary Collapse

Collapse condition:

$$\Delta B < 0.$$

Collapse velocity:

$$V_c = -\frac{dB}{dt}.$$

6 Boundary Drift

Drift velocity:

$$V_d = \frac{dB}{dt}.$$

7 Algebraic Motif Interaction

Interaction strength:

$$A = k(M_1, M_2, B).$$

8 Boundary Regime Transitions

Transition strength:

$$T = q(B, R).$$

9 Conclusion

Boundary Motif Algebra defines alignment, interference, coupling, collapse, drift, and algebraic interaction at boundaries. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.